Assignment 2

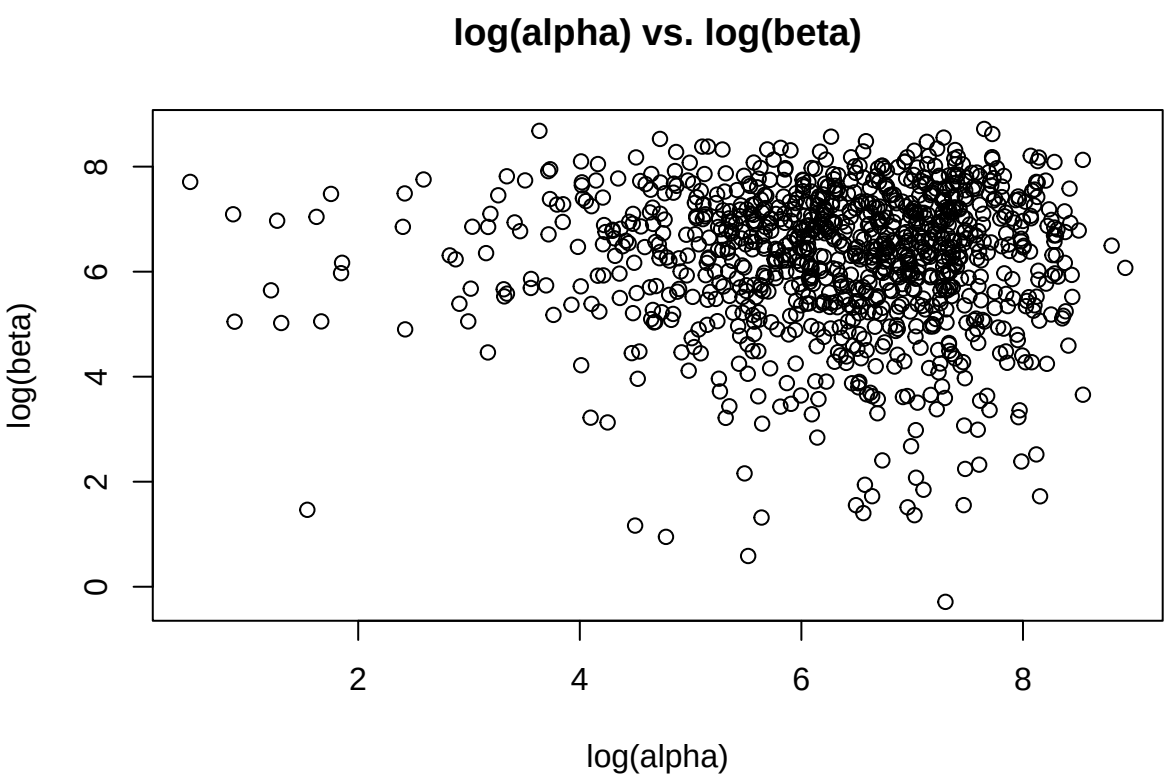
Problem 1

a.

$$\theta_j \mid \alpha, \beta \sim Beta(\alpha, \beta)$$
$$\alpha, \beta \sim iid \quad Expon(0.001)$$

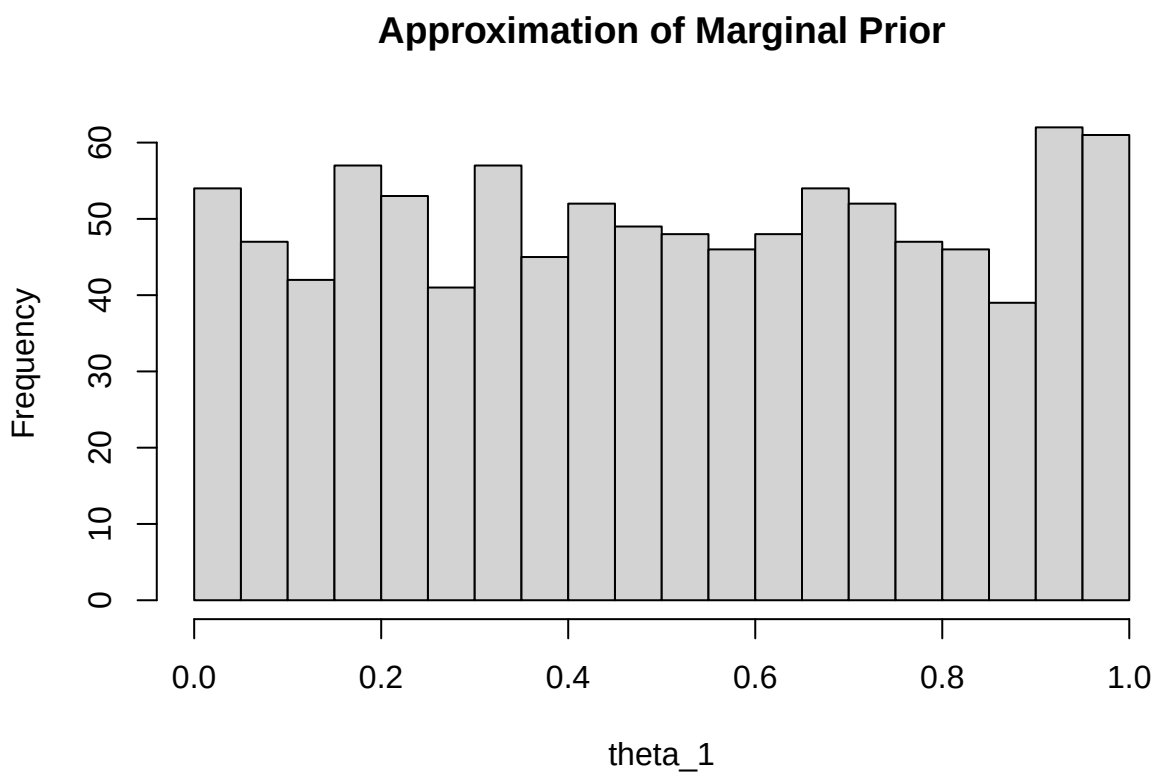
a.i. Simulate 1000 (α, β) pairs and produce a scatter plot of $log(\alpha)$ and $log(\beta)$

```
alpha <- rexp(1000, 0.001)
beta <- rexp(1000, 0.001)
plot(
    log(alpha),
    log(beta),
    main="log(alpha) vs. log(beta)",
    xlab="log(alpha)",
    ylab="log(beta)"
)
```



a.ii. Forward simulate θ_1 and produce a histogram

```
theta_1 <- rbeta(1000, alpha, beta)
hist(theta_1, breaks = 30, main="Approximation of Marginal Prior")
```



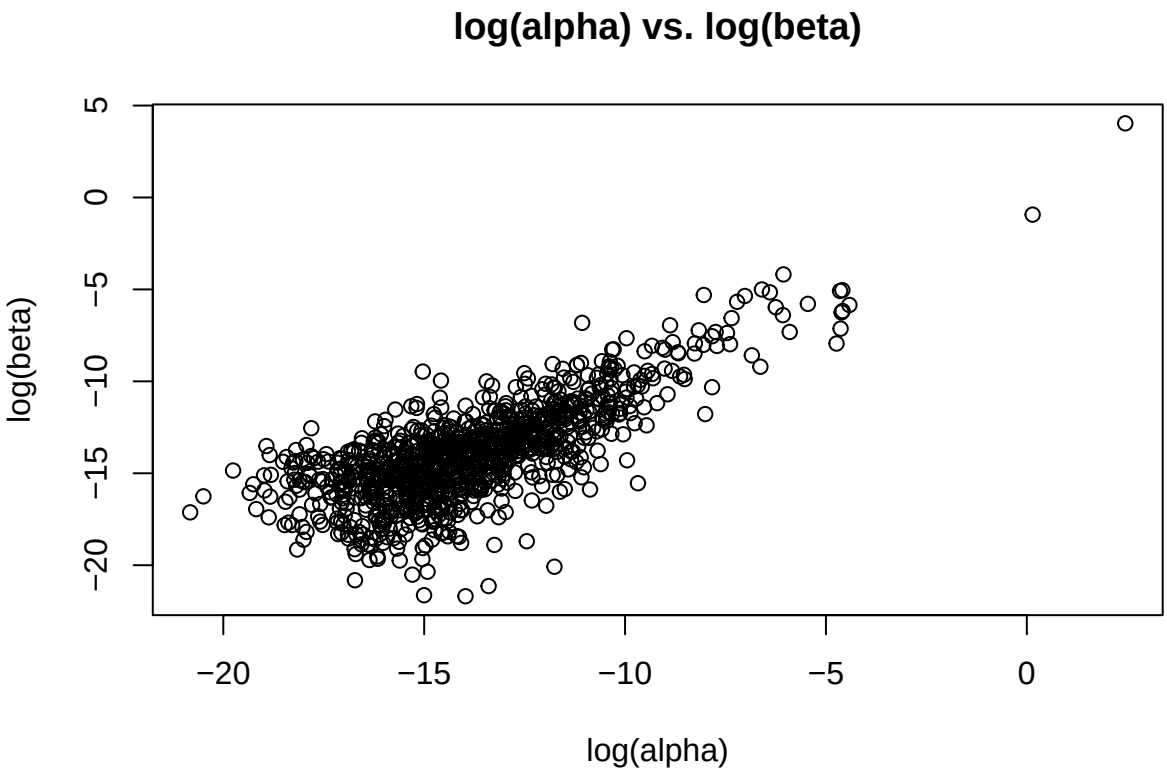
b.

$$\theta_j \mid \alpha, \beta \sim Beta(\alpha, \beta)$$
$$\alpha = \frac{\phi_1}{\phi_2^2} \quad \beta = \frac{(1-\phi_1)}{\phi_2^2}$$
$$\phi_1 \sim U(0,1) \quad \phi_2 \sim Expon(0.001)$$

b.i. Independently simulate 1000 pairs (α, β) from their hyperprior and produce a scatter plot of $log(\alpha)$ and $log(\beta)$

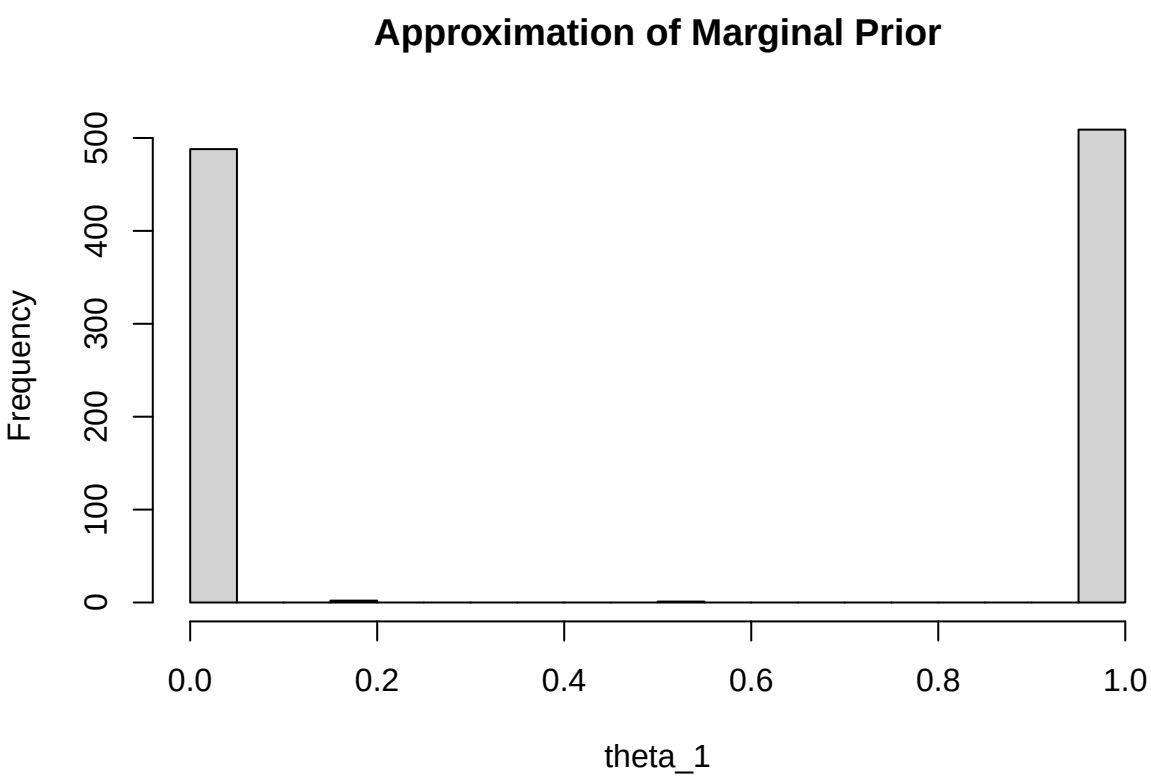
```
phi_1 <- runif(1000)
phi_2 <- rexp(1000, 0.001)
alpha <- phi_1 / phi_2^2
beta <- (1 - phi_1) / phi_2^2

plot(
    log(alpha),
    log(beta),
    main="log(alpha) vs. log(beta)",
    xlab="log(alpha)",
    ylab="log(beta)"
)
```



b.ii. Forward simulate θ_1 and produce a histogram

```
theta_1 <- rbeta(1000, alpha, beta)
hist(theta_1, breaks = 30, main="Approximation of Marginal Prior")
```



Problem 2

$$
\begin{aligned}
&\hat{\psi}_j \mid \psi_j \sim \quad indep. \;\; N(\psi_j, \sigma_j^2) \quad j = 1, \ldots, 12 \\
&\psi_j \mid \psi_0, \sigma_0 \sim \quad iid \;\; N(\psi_0, \sigma_0^2) \quad j = 1, \ldots, 12 \\
&\psi_0 \sim N(0, 1000^2) \\
&\sigma_0 \sim U(0, 1000)
\end{aligned}
$$

where

- ψ_0 and σ_0 are independent.
- $\sigma_j,\ j = 1, \ldots, 12$ are fixed and known.

a. For each hyperparameter, specify an improper density

hyperparamters

- σ_0
- ψ_0

Improper densities

$N(0, 1000^2)$

- The hyperprior of ψ_0, $N(0, 1000^2)$, is rather disperse given the variance 1000^2.
- Apparently, it approximates $\psi_0 \propto 1 \quad \psi_0 \in \mathbb{R}$, an improper uniform prior on the real line.

$U(0, 1000)$

- The hyperprior of σ_0, $U(0, 1000)$ is rather disperse as its upper bound is rather large.
- Apparently, it approximates $\sigma_0 \propto 1 \quad \sigma_0 > 0$, an improper uniform prior on the real line.

b. Draw a DAG for the model

c. Show the JAGS code

```
model {
  for (j in 1:length(psihat)) {
    psihat[j] ~ dnorm(psi[j], 1/(se[j]^2))
    psi[j] ~ dnorm(psi0, 1/sigmasq0)
  }

  psi0 ~ dnorm(0, 0.000001)
  sigma0 ~ dunif(0, 1000)

  sigmasq0 <- sigma0^2
}
```

d. Set up the rjag model

```r
library("rjags")
d <- read.table("./Heart Studies.txt")
colnames(d) <- c("i", "psihat", "se")
m <- jags.model("./asgn2template.bug", d)
```

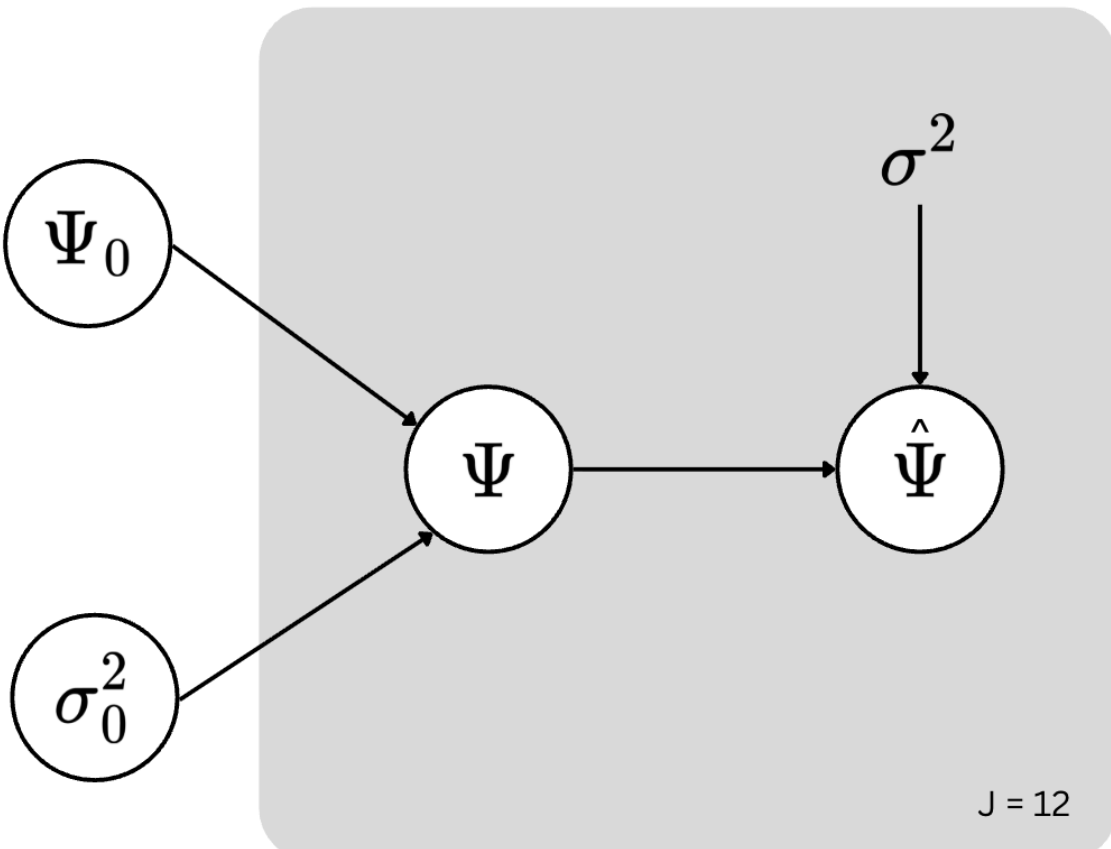


Figure 1: dag image

```
## Warning in jags.model("./asgn2template.bug", d): Unused
## variable "i" in data
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 12
##    Unobserved stochastic nodes: 14
##    Total graph size: 69
##
## Initializing model
##
##   |                                                              |
```

```
print(d)
```

```
##     i psihat    se
## 1   1  1.055 0.373
## 2   2 -0.097 0.116
## 3   3  0.626 0.229
## 4   4  0.017 0.117
## 5   5  1.068 0.471
## 6   6 -0.025 0.120
## 7   7 -0.117 0.220
## 8   8 -0.381 0.239
## 9   9  0.507 0.186
## 10 10  0.000 0.328
## 11 11  0.385 0.206
## 12 12  0.405 0.254
```

e. Run the model and retrieve numerical summaries for the posterior ψ_0 and σ_0^2

```
update(m, 10000)
```

```
##   |                                                              |
```

```
x <- coda.samples(m, c("psi0", "sigmasq0"), 100000)
```

```
##   |                                                              |
```

```
psi0 <- as.matrix(x)[, "psi0"]
sigmasq0 <- as.matrix(x)[, "sigmasq0"]
subset(summary(x)$statistics, select = c("Mean", "SD"))
```

```
##               Mean        SD
## psi0     0.2143380 0.1332024
## sigmasq0 0.1499912 0.1319621
```

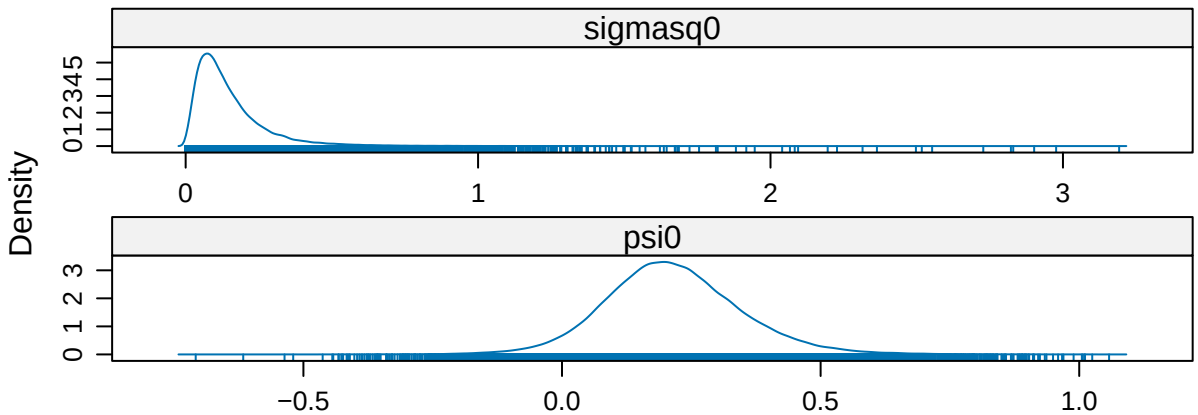
```
quantile(psi0, c(0.025, 0.975))
```

```
##        2.5%       97.5%
## -0.03263315  0.49566311
```

```
quantile(sigmasq0, c(0.025, 0.975))
```

```
##       2.5%      97.5%
## 0.01864988 0.48561568
```

```r
require("lattice")
densityplot(x[, c("psi0", "sigmasq0")])
```



e.i. The posterior numerical summary of ψ_0

- **Expected Value**: 0.2146998
- **Standard Deviation**: 0.1326329
- **95% Central Interval**: $[-0.032659950.49480561]$

e.ii.The posterior numerical summary of σ_0^2

- **Expected Value**: 0.1508600
- **Standard Deviation**: 0.1363831

- **95% Central Interval**: $[0.01883460.4858067]$

f. With newly added case-control study

f.i. Draw the updated DAG

- Since the newly added study are conditionally independent of previous studies give ψ_0 and σ_0, we can modify the DAG as follows:

where

- σ_{13}: The log-odds standard error of the new study ($\sigma_{13} = 0.4$)
- $\hat{\psi}_{13}$: The new estimated logs-odds ratio
- ψ_{13}: The new logs-odds ratio

f.ii. Modified JAGS Model

```
model {
  for (j in 1:length(psihat)) {
    psihat[j] ~ dnorm(psi[j], 1/(se[j]^2))
    psi[j] ~ dnorm(psi0, 1/sigmasq0)
  }

  psi0 ~ dnorm(0, 0.000001)
  sigma0 ~ dunif(0, 1000)
  sigmasq0 <- sigma0^2

  psi_13 ~ dnorm(psi0, 1/sigmasq0)
```

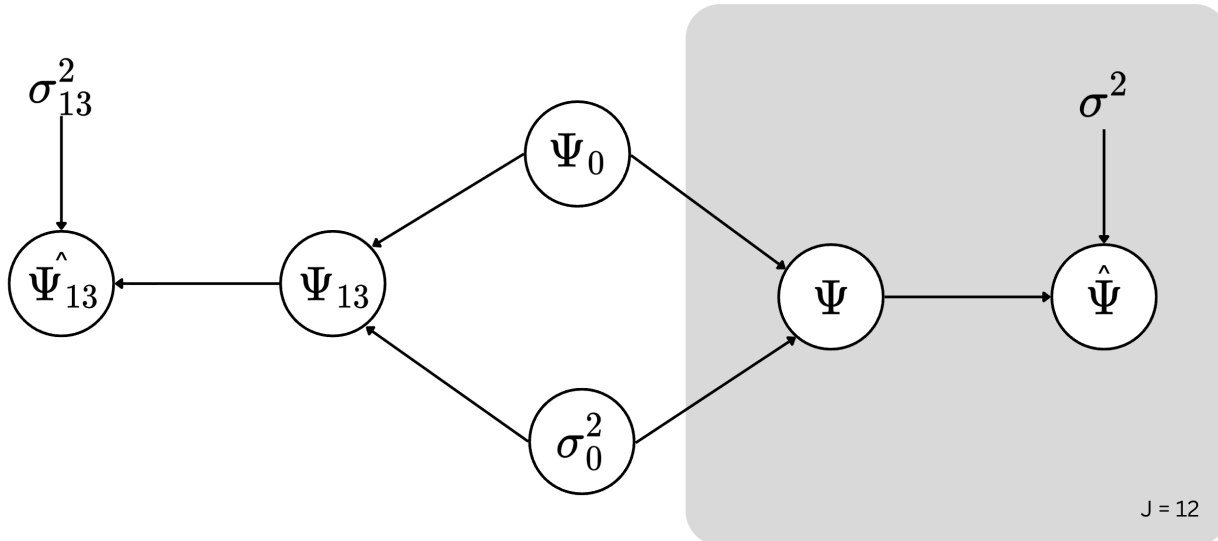


Figure 2: Update Dag

```
  psihat_13 ~ dnorm(psi_13, 6.25)
}
```

f.iii. Retrieve numerical summaries for the posterior $\hat{\psi}_{13}$

```
m <- jags.model("./updated_asgn2template.bug", d)
```

```
## Warning in jags.model("./updated_asgn2template.bug", d):
## Unused variable "i" in data
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 12
##    Unobserved stochastic nodes: 16
##    Total graph size: 72
##
## Initializing model
##
##   |                                                              |
```

```
update(m, 10000)
```

```
##   |                                                              |
```

```
x <- coda.samples(m, c("psihat_13"), 100000)
```

```
##   |                                                              |
```

```
summary(x)$statistics[c("Mean", "SD")]
```

```
##      Mean        SD
```

```
## 0.2146031 0.5734125
```

```r
psihat_13 <- as.matrix(x)[, "psihat_13"]
quantile(psihat_13, c(0.025, 0.975))
```

```
##       2.5%      97.5%
## -0.9067015  1.3612881
```

- **Expected Value**: 0.2131864
- **Standard Deviation**: 0.5690035
- **95% Central Interval**: $[-0.89871151.3548126]$

f.iv. Add an indicator function to JAGS Model

```
model {
  for (j in 1:length(psihat)) {
    psihat[j] ~ dnorm(psi[j], 1/(se[j]^2))
    psi[j] ~ dnorm(psi0, 1/sigmasq0)
  }

  psi0 ~ dnorm(0, 0.000001)
  sigma0 ~ dunif(0, 1000)
  sigmasq0 <- sigma0^2

  psi_13 ~ dnorm(psi0, 1/sigmasq0)
  psihat_13 ~ dnorm(psi_13, 6.25)
  ind <- psihat_13 > 0.8
}
```

```r
m <- jags.model("./indicator_asgn2template.bug", d)
```

```
## Warning in jags.model("./indicator_asgn2template.bug",
## d): Unused variable "i" in data
```

```
## Compiling model graph
##    Resolving undeclared variables
##    Allocating nodes
## Graph information:
##    Observed stochastic nodes: 12
##    Unobserved stochastic nodes: 16
##    Total graph size: 74
##
## Initializing model
##
##   |                                                              |
```

```r
update(m, 10000)
```

```
##   |                                                              |
```

```r
x <- coda.samples(m, c("ind"), 100000)
```

```
##   |                                                              |
```

```r
summary(x)$statistics["Mean"]
```

```
##    Mean
## 0.14436
```